$$S=\int d^4x\,\sqrt{-g}\Big[\frac{1}{2}M_{\rm Pl}^2R+M_{\rm Pl}^2\dot{H}g^{00}-M_{\rm Pl}^2(3H^2+\dot{H})+\frac{1}{2!}M_2(t)^4(\delta g^{00})^2+\frac{1}{3!}M_3(t)^4(\delta g^{00})^3+$$
$$-\frac{\bar{M}_1(t)^3}{2}(\delta g^{00})\delta K^\mu{}_\mu-\frac{\bar{M}_2(t)^2}{2}\delta K^\mu{}_\mu{}^2-\frac{\bar{M}_3(t)^2}{2}\delta K^\mu{}_\nu\delta K^\nu{}_\mu+...\Big]\,. \tag{163}$$